Web Services, Web API, REST, MVC

SWE 4537: Server Programming
Lecture 2

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05 November 2026

Recap: Where We Left Off (Lecture 1)

The Core Loop

In our last lecture, we established the fundamentals of web communication.

- A browser (Client) sends an HTTP Request to a server at a specific IP and Port.
- The Server then processes the request and sends back an HTTP Response (like 200 `OK` + HTML).

The Bridge to Today

Today's Focus

What is the actual program on the server that is responsible for:

- 👉 Listening for those requests?
- 👉 Deciding what to do with them?
- 👉 Creating that response?

The Answer

This program, in its broadest sense, is a Web Service.

Key Distinction: App vs. Service

Web Application (For Humans)

A program whose primary "user" is a human, and its output is a User Interface (UI).

- **Analogy:** A fully-assembled car.
- **Response:** Sends HTML, CSS, and JavaScript for a browser to render visually.

Web Service (For Machines)

A program whose primary "user" is another program (a machine), and its output is raw data.

- **Analogy:** The engine from the car factory.
- **Response:** Sends JSON (or XML) for another program to parse and use.

Paradigms of Web Services

How do machines talk to each other?

RPC (Remote Procedure Call)

The oldest idea. "Just call a function on another server."

- Client code calls `getUsers()` or `calculateTotal(orderID)`.
- It's simple, but brittle. What if the server renames the function? How do you discover functions?

SOAP (Simple Object Access Protocol)

A formal protocol (a strict set of rules) for web services.

- Uses complex XML for all messages.
- Has a rigid contract (WSDL) defining all functions.
- Very powerful (built-in security, transactions) but very heavyweight and complex.

What is WSDL? (Part I)

Web Services Description Language

The "Contract"

An XML-based file that acts as a formal "contract" or "user manual" for a web service.
It tells a client application *exactly* how to structure a request and what to expect in response.

WSDL vs. SOAP

- 👉 **WSDL** is the description (the "menu").
- 👉 **SOAP** is the message format (the "order").

You read the WSDL to learn how to build the SOAP message.

What is WSDL? (Part II)

Key Components:

A WSDL file answers these 5 questions:

- 👉 <types>: What are the data types?
- 👉 <message>: What do the payloads look like?
- 👉 <portType>: What are the abstract operations?
- 👉 <binding>: What protocol do we use? (e.g., SOAP)
- 👉 <service>: Where is the service's URL?

What is WSDL? (Part III)

Simplified WSDL Example

```
<!-- 1. The data types we will use --> <types> <schema> <element name="UserID" type="xs:int"/> <element
name="UserName" type="xs:string"/> </schema> </types>
<!-- 2. The request/response messages --> <message name="getUserRequest"> <part name="id"
element="tns:UserID"/> </message> <message name="getUserResponse"> <part name="name"
element="tns:UserName"/> </message>
<!-- 3. The abstract operations (functions) --> <portType name="UserPortType"> <operation name="getUser">
<input message="tns:getUserRequest"/> <output message="tns:getUserResponse"/> </operation> </portType>
<!-- 4. The concrete protocol (SOAP) --> <binding name="UserBinding" type="tns:UserPortType"> <soap:binding
style="rpc" ... /> <operation name="getUser"> <soap:operation soapAction="getUser"/> <input><soap:body
.../></input> <output><soap:body .../></output> </operation> </binding>
<!-- 5. The physical address (URL) --> <service name="UserService"> <port name="UserPort"
binding="tns:UserBinding"> <soap:address location="http://api.example.com/UserService"/> </port> </service>
</definitions>
```

What is an API?

API: Application Programming Interface

This is the "public-facing" part of the server. It's the contract that defines *how* other programs can talk to your service.

- 👉 It's the "waiter" for the kitchen.
- 👉 It's the "menu" that lists the rules.
- 👉 It's the technical specification manual.

The Restaurant Analogy

- 👤 **The Client (You):** You want food (data).
- 🔪 **The Server (The Kitchen):** Has the data (ingredients) and the logic (chefs) to make your food. This is the Web Service.
- 🤷 **The API (The Waiter):** You (the client) are not allowed to go into the kitchen. You must talk to the waiter.

Connecting the Analogy to Theory

How do different paradigms use the "waiter"?

RPC Analogy

You shout into the kitchen: `"getMeSteak()"`.
Problem: But what if the chef's function is named `fetchSteak(temp)`?

SOAP Analogy

The waiter gives you a legal document (the WSDL) in Latin (XML). You must fill out a specific form *exactly* to order.
Problem: It's very precise, but extremely slow and difficult to work with.

REST Analogy

The waiter gives you a simple, well-organized menu (the API endpoints). You use simple words (Verbs) to pick items (Nouns). "I'll `GET` the `steak`."

What is REST?

REST (Representational State Transfer)

An architectural style. It's a philosophy defined by a set of constraints that, if followed, lead to a scalable, evolvable system.

The 6 Guiding Constraints

1. **Client-Server:** Enforces Separation of Concerns.
2. **Stateless:** (The most important one! More soon.)
3. **Cacheable:** Responses can be cached by the client.
4. **Uniform Interface:** (The key to REST!)
5. **Layered System:** Proxies/gateways can exist in the middle.
6. **Code on Demand (Optional):** Server can send code (e.g., JavaScript).

REST vs. RPC: A Shift in Thinking

This is the core of the "Uniform Interface" constraint.

RPC (Remote Procedure Call)

It's about Actions (Verbs).

- Endpoints are functions: `/getAllTodos`, `/createNewTodo`.
- **Problem:** This is brittle. What if you need todos by date? You must add a *new function* and *new endpoint*: `/getAllTodosByDate`.

REST (Representational State Transfer)

It's about Resources (Nouns).

- You have one resource: `/todos`.
- You use parameters for filtering: `/todos?date=2025-11-13`.
- You use HTTP Verbs (the existing protocol!) for actions.

The RESTful "Menu" (Part 1: The Collection)

Resource: `/todos`

(Refers to the collection of ALL todos)

👉 `GET /todos`

- **Action:** Read
- **Meaning:** "Get me a list of all todos."
- **Note:** Data (like filters) is sent in the URL (e.g., `/todos?user=1`).

👉 `POST /todos` →

- **Action:** Create
- **Meaning:** "Create a new todo." (Data is in the body).
- **Note:** Data is in the body, not visible in the URL. Good for sensitive data.

The RESTful "Menu" (Part 2: A Specific Item)

Resource: `/todos/123`

(Refers to a single, specific todo)

- `GET /todos/123` →**Read**: "Get details for todo #123."
- `PUT /todos/123` →**Update**: "Replace todo #123."
- `DELETE /todos/123` →**Delete**: "Delete todo #123."

Note

This is clean, predictable, and easy to understand. We use the built-in power of HTTP instead of inventing new URLs.

More HTTP Methods

HEAD

Identical to `GET`, but it does not return the message body

👉 **Use:** Check if a resource exists or get its metadata (like `Content-Length`) without downloading the whole file.

PATCH

Applies a partial replacement to a resource

👉 **Use:** Unlike `PUT` (which replaces everything), `PATCH` is for changing just one field (e.g., updating *only* a todo's title).

OPTIONS

Asks the server to list the supported HTTP methods for a given URL

👉 **Use:** Used to describe the communication options for the target resource, such as the allowed HTTP methods.

REST: The "Representation" Part

The "Representation" in REST

The server doesn't send you the actual database object. It sends a *representation* of that resource.

- 👉 `GET /todos/123` → JSON (for a mobile app)
- 👉 `GET /todos/123` → HTML (for a website)

Content Negotiation

This is formally called Content Negotiation.

- 👉 Client sends: `Accept: application/json`
- 👉 Server replies: `Content-Type: application/json`

For this course, we will *always* use JSON.

REST's Most Important Constraint: Statelessness

Stateless

The server does not store any client "session" state between requests.

What this means:

Every single request from the client must contain all the information needed for the server to process it.

- **Bad (Stateful):** 1. `POST /login` 2. `GET /my-todos` (Server just "remembers" who you are).
- **Good (Stateless):** 1. `POST /login` (Server returns an `access_token`). 2. `GET /todos` (Client sends `Authorization: Bearer <token>` in the header).

Why?

Scalability. Any server in a 1000-server cluster can handle any request, as no server needs to remember "who you are."

The Problem: "The Big Messy File"

The Naive Approach

"Okay, so we're going to build a program that listens for these requests. We'll use a framework like `FastAPI`."

The Temptation

"In Week 2, our `main.py` file will look like this..."

The "Messy" File Example

main.py

```
# main.py
@app.get("/todos")
def get_all_todos():
    # 1. Connect to the database...
    # 2. Write a SQL query...
    # 3. Format the data into JSON...
    # 4. Return the JSON...

@app.post("/todos")
def create_todo():
    # 1. Get the JSON from the request body...
    # 2. Connect to the database...
    # 3. Write a SQL 'INSERT' query...
    # 4. Return a '201 Created' status...
```

The Problem: "The Big Messy File"

The Problem

What happens when we add `/users`, `/posts`, `/comments`, and all their `GET`, `POST`, `PUT`, `DELETE` methods?

- 👉 Our `main.py` file becomes 5,000 lines long!
- 👉 All our database code, business logic, and API code are mixed together.
- 👉 It's impossible to test, debug, or maintain.

This violates a core software principle:

Separation of Concerns

The Solution: Architectural Patterns

Architectural Pattern

A template for organizing your code. A reusable solution to a common problem.

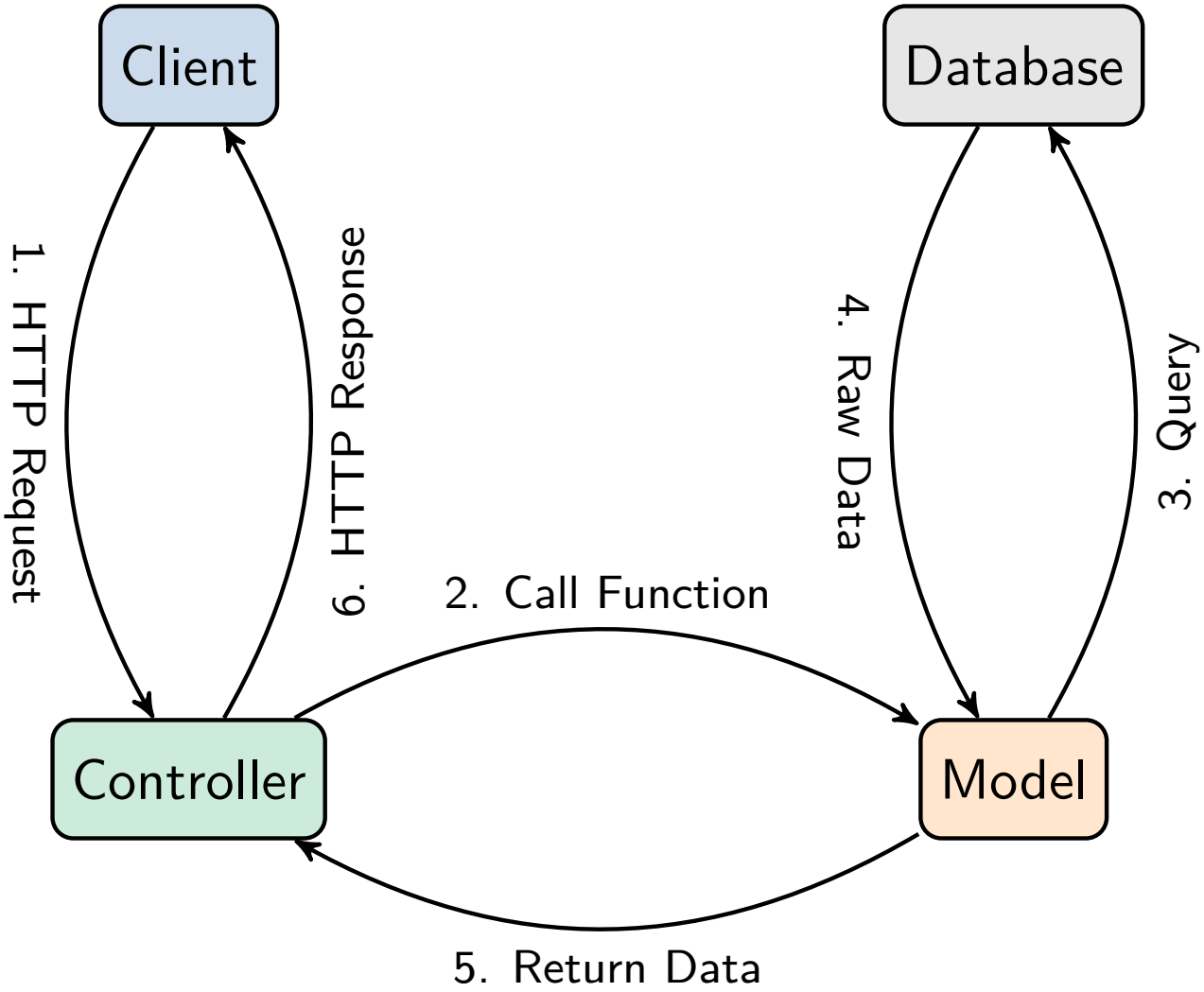
Our Pattern: MVC

The most famous pattern is MVC (Model-View-Controller). We will use a variation of this for our API.

Core Idea

Separation of Concerns. We separate our code into three distinct jobs.

The MVC Pattern (Our API Flow)



MVC: The Components (M)

M - Model (The "Data" Layer)

This is the "database" layer. It handles all data and business logic.

- Its *only* job is to talk to the database (e.g., `SELECT * from todos`).
- It contains all your SQL queries.
- It enforces rules (e.g., "password must be 8 characters").
- It doesn't know the API exists. It just returns data.

MVC: The Components (C)

C - Controller (The "Brain" / "Waiter")

This is the "brain" of the operation. It's the "traffic cop" that directs flow.

- It handles the HTTP Request (e.g., `GET /todos`).
- It calls the Model ("Hey Model, get me all the todos").
- The Model gives it raw data.
- The Controller then formats that data into a Response (the "View").

MVC: The Components (V)

V - View (The "Presentation" Layer)

Formats the data for the "user".

- **In a traditional website:** This is the HTML file / template.
- **For our API:** The "user" is a machine. The "View" is simply the JSON that the Controller sends back.

Our New, Clean Flow (MVC)

This is the same process, but organized:

1. Request for `GET /todos` hits the Controller.
2. Controller calls the Model (`model.get_all_todos()`).
3. Model queries the Database and returns raw data.
4. Controller formats the data into JSON (the "View").
5. Controller sends the `200 OK` Response.

Why is this better?

If we change the database? Only touch the Model.
If we change the URL? Only touch the Controller.

We Have All The Theory!

We now have all the theory we need.

- 👉 We know the rules (HTTP).
- 👉 We know the design (REST).
- 👉 We know the organization (MVC).

Questions?

Thank You!

Thank you for your time and attention.

Any questions or clarifications?



